

Samita

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Professional Summary

Software Development Engineer with experience building full-stack web applications with responsive web design and cross-platform mobile applications. Skilled in React, React Native (Expo), Node.js/Express, Next.js, MongoDB, and native Android (Kotlin) development. Strong background in REST API design, native module bridging, database modeling, and shipping production mobile builds via CI/CD.

Professional Experience

Samsara Wellness

Software Development Engineer

Dec 2025 – June 2026

Bengaluru, India

Admin Dashboard – Membership & Health CRM System (MERN)

- Built an admin dashboard using Next.js (TypeScript) to manage users, memberships, and health data at scale.
- Developed membership APIs in Node.js + Express + MongoDB for trial plans, lifetime plans, coupon-based access, renewals, cancellations, and expiry tracking.
- Created a separate admin service layer to fetch per-user active memberships and history, integrating it into a 100+ user listing table.
- Implemented period tracking & prediction features, including cycle history, phase detection, and upcoming period forecasts.
- Integrated secure authenticated API calls, handled async parallel data fetching, and resolved route mismatches, ES module export issues, 404 errors, and frontend-backend field-mapping bugs.
- Debugged real-world production issues involving API responses, schema mismatches, date handling, and multi-user state synchronization.

HealthSync – Health Connect / HealthKit Dashboard (React Native, Expo, Kotlin, TypeScript)

- Built a cross-platform (Android & iOS) app aggregating steps, heart rate, calories, and sleep from Android Health Connect and Apple HealthKit into a single dashboard.
- Developed a custom native Kotlin module (via React Native's New Architecture) to bridge Health Connect on Android, including manual AndroidManifest.xml configuration (activity-alias, permission rationale activity) for native in-app permission flows on Android 13/14+.
- Implemented platform-specific data-fetching logic with a unified dispatcher – querying Health Connect record types on Android and HealthKit quantity/category samples (via Nitro Modules) on iOS, including Apple Watch-sourced data.
- Configured Expo config plugins (expo-build-properties, react-native-health-connect, @kingstinct/react-native-healthkit) and set up EAS Build with platform-specific build profiles for Android APK and iOS internal distribution.

PulseOnly – Camera-Based Heart Rate (PPG) Monitor (React Native, Kotlin, Android CameraX)

- Built an Android app measuring heart rate (BPM) using only the phone's rear camera and flash (photoplethysmography), pairing a React Native UI (navigation, live progress bar, permission handling) with a custom native processing module.
- Wrote a custom native Kotlin module (PpgModule, registered via a custom ReactPackage) using CameraX with Camera2 interop to lock auto-exposure and enable torch, streaming YUV camera

frames for background analysis.

- Implemented a real-time signal-processing pipeline in Kotlin: YUV→RGB conversion, time-series re-sampling, a 2nd-order Butterworth bandpass filter (0.7–4 Hz), signal quality scoring, autocorrelation-based peak detection, and BPM calculation from the dominant lag.
- Bridged native events (onStarted, onProgress, onResult, onError) to the JS layer via NativeEventEmitter, driving a live scanning UI with progress %, error alerts, and BPM display.

Samsara Wellness

Software Development Engineer Intern

Sept 2025 – Nov 2025

Bengaluru, India

PulseApp – Multi-Feature Health & Wellness App (React Native, Expo, React Navigation, REST APIs)

- Built a cross-platform app with tab-style stack navigation covering pulse/heart-rate scan, an Ayurveda chatbot, and barcode/food-label scanners.
- Implemented a barcode scanner screen (expo-image-picker, expo-image-manipulator) that compresses and POSTs images to a remote API to retrieve product name, nutrition facts, allergens, and veg/non-veg classification.
- Implemented a food scanner screen uploading images as multipart form data to a separate analysis API, rendering per-item macros (carbs, fat, protein, fiber) and computed totals.
- Added local persistence with AsyncStorage to cache the last 50 scanned foods, with automatic expiry/cleanup of entries older than 24 hours.
- Built a chat UI sending message history to a hosted chatbot endpoint, rendering a WhatsApp-style conversation with typing indicator and auto-scroll; custom-themed the app via react-native-paper and React Navigation theme overrides.

Inonetecx Inc.

Machine Learning & Web Development Intern

Dec 2024 – Mar 2025 (Remote)

Toronto, Canada

- Developed and maintained a Flask-based web application, contributing to backend logic, feature workflows, and modular service design.
- Implemented a complete user wishlist system, allowing users to create wishlists, add products, and manage saved items with seamless UI integration.
- Built an automated service that fetches and extracts product details from shared product links using Selenium for text data scraping, reducing manual input and improving user convenience.

Projects

ML/DL-Based Yoga Pose Recognition System

Patented

YOLOv10, OpenCV, MediaPipe

- Built a real-time posture-alignment system using YOLOv10 Nano and 17-point body keypoint detection; curated and annotated a 2,500–3,000 image dataset (Roboflow), achieving 92% accuracy, 88% precision, and 91% recall.
- Awarded Indian Patent No. 202511067136A for the posture monitoring and alignment-correction approach.

Education

Quantum University, Roorkee

B.Tech in Computer Science & Engineering (Data Science)

2022 – 2026

CGPA: 8.13/10

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, SQL, C++, HTML, CSS, HTML5, CSS3

- **Frameworks & Libraries:** React.js, React Native (Expo), Node.js, Express.js, Next.js, Spring Boot, Bootstrap
- **Databases:** MongoDB, PostgreSQL, MySQL, NoSQL
- **Mobile / Native:** Kotlin, Android CameraX, Health Connect, Apple HealthKit, Nitro Modules
- **Tools & Platforms:** Git, GitHub, VS Code, Linux, EAS Build (CI/CD), REST APIs, Bash

Certifications & Achievements

- The Complete Python Bootcamp – Python 3, OOP & Flask (Udemy)
- Machine Learning A-Z: AI, Python & Data Science (Udemy)
- Python and Flask Bootcamp – REST APIs with Flask (Udemy)
- Patented Yoga Posture Monitoring System (Patent No. 202511067136A)